Geophys. J. Int. (2001) **147,** 294–309

A palaeomagnetic study and age determinations of Tertiary rocks in Nicaragua, Central America

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Accepted 2001 May 31. Received 2001 May 30; in original form 2000 February 4

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SUMMARY

Palaeomagnetic, K–Ar and Ar^{40}/Ar^{39} measurements have been made to elucidate Nicaragua's plate tectonic history. These show that the Nicaraguan Highland rotated 90° counterclockwise between 30 and 18 Myr ago, but the Pacific Coastal Plain has not rotated.

A lack of data has prevented a direct comparison with the tectonic history of adjacent areas, in particular, the Chortis Block, which is the part of the Caribbean Plate that is geologically most closely related to the Nicaraguan Highland. However, our data suggest that the Caribbean Plate in this part of Central America was not consolidated until *c.* 15 Ma.

Key words: Nicaragua, palaeomagnetism, radiometric ages, tectonic reconstructions, Tertiary rocks.

INTRODUCTION

Nicaragua forms part of the Caribbean Plate (Fig. 1), which is built up of a number of allochthonous terranes (Case *et al.* 1984) and has a complex geological history resulting from the interaction between the North and South American, Farallon, Cocos and Nazca plates. The present boundaries of the Caribbean Plate are defined by the distribution of seismic activity (Molnar & Sykes 1969). Former plate boundaries evolved from the relative positions of North and South America following the Late Triassic–Jurassic break-up of Pangaea. In most Pangaea reconstructions the area presently occupied by the Caribbean Plate was overlapped by North and South America (Bullard *et al.* 1965; Van der Voo *et al.* 1976; Pindell 1985), and the Caribbean Plate is regarded as allochthonous to its neighbouring plates. This means that North and South America are not the rifted margins of the present-day Caribbean basin.

Duncan & Hargraves (1984) presented a model for the evolution of the Caribbean in which the core of the plate was formed from the lava that was poured out onto the Farallon Plate when it moved eastwards over the Galapagos hotspot. When the space between the Americas was big enough the Caribbean entered the gap. At *c.* 80 Ma the migration of South America away from North America was more or less completed (Pindell 1985) and this may have been the time that the Caribbean Plate entered.

Caribbean Plate boundaries have been predominantly either strike-slip or convergent. Pindell & Dewey (1982) and Sykes *et al.* (1982) suggest two distinct stages for the development of the northern Caribbean Plate boundary. The first is characterized by a Late Cretaceous–Palaeocene northward convergence, resulting in collision and suturing of the Antillean arc with the Bahamas platform. This stage ended during the latest Early Eocene–earliest Middle Eocene. The Caribbean Plate then started to move eastwards relative to the North and South American plates. The western part of this strike-slip boundary evolved as a large pull-apart basin, the Cayman Trough. The time of opening of the Cayman Trough has been estimated at 45–50 Ma (Rosencrantz *et al.* 1988), and, from magnetic anomalies produced by sea-floor spreading in the trough, the rate of eastward movement has been estimated (Rosencrantz *et al.* 1988). The trough opened at a rate of 20–30 mm yr^{-1} from *c.* 50 to 30 Ma, when there was shift in the spreading ridges and a slow-down of opening to less than 15 mm yr^{-1}. This probably reflects a reorganization of relative motion between the North America and Caribbean plates.

As the Caribbean Plate is composed of various blocks and microplates, a tectonic reconstruction of these is a basis for the understanding of the formation of the Caribbean. Central America can be divided into four major regions (Dengo 1969, 1985; Escalante 1990), the Maya, Chortis, Chorotega and Choco Blocks (Fig. 1). The Chortis Block is defined to encompass southern Guatemala, Honduras, northern Nicaragua and the western Nicaraguan Rise. This block has a pre-Mesozoic continental-type crust, unlike the oceanic Chorotega Block in the south. Central Nicaragua has been related to neither the Chortis Block nor the Chorotega Block, while southwestern Nicaragua may form part of the Chorotega Block (Di Marco *et al.* 1995). The Chorotega Block constitutes most of the southern Middle American landbridge and it has been suggested that it formed the western edge of the Caribbean Plate during the Late Cretaceous–Palaeocene (Di Marco *et al.* 1995). In an

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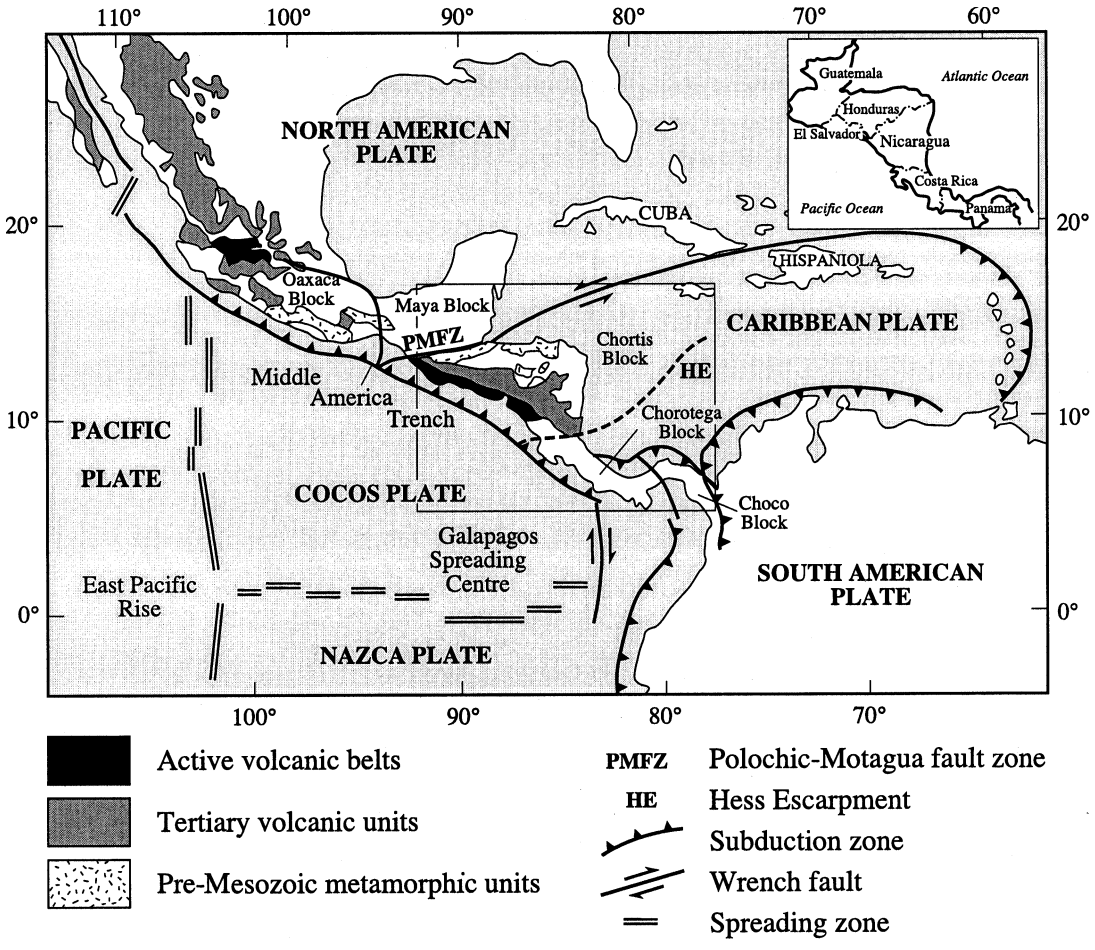


Figure 1. Regional geology of Central America and major crustal blocks.

attempt to understand Nicaragua's plate tectonic context (Elming 1983, 1985, 1992), our palaeomagnetic study was performed together with K–Ar and $^{40}Ar/^{39}Ar$ age determinations on predominantly Tertiary volcanic rocks from the central and western parts of the country.

GEOLOGY OF NICARAGUA

Nicaragua can be divided into at least two different tectonic regimes separated by the SE–NW-trending Nicaraguan Depression (Fig. 2), which has been defined as a half-graben (McBirney & Williams 1965). The Cocos Plate subducts below the Caribbean Plate with northeastward motion, and an associated chain of volcanoes is located in the western edge of the depression parallel to the trench.

East of the depression, Nicaragua is dominated by Tertiary volcanic rocks of the Central American Highlands, which extend SE–NW through the country. In northern Nicaragua these volcanic rocks are superimposed on metamorphosed Palaeozoic and Mesozoic rock formations. The volcanic rocks can be divided into the Matagalpa and Coyol groups. The Oligocene–Palaeocene Matagalpa Group volcanics are thought to be deposited on an ignimbrite plateau (Ehrenborg 1996). The Pliocene–Miocene Coyol Group, which consists mainly of lavas, breccias, lahars and locally of pyroclastic rocks, has been divided into upper and lower subgroups (Parsons Corporation Report 1972), characterized by basaltic rocks overlain by intermediate to acid volcanic rocks. Ehrenborg (1996) has explained the repeated stratigraphic sequence using a facies model, relating various volcanic centres to each other and superseding the older subdivision of the Coyol Group.

To the west of the depression, in the Pacific Coastal Plain, the rocks are predominantly of sedimentary origin and are assumed to have been deposited on a Nicoya Complex type of basement (Parsons Corporation Report 1972). The sediments, tuffaceous shale, siltstone, greywacke, sandstone and limestone, range in age from Cretaceous to Miocene (Rivas, Brito, Masachapa and El Fraile formations) and are overlain by sediments of similar composition to the El Salto Formation and the pyroclastic deposits of the La Sierra Group (McBirney & Williams 1965; Parsons Corporation Report 1972).

Seyfried *et al.* (1987) regard the Upper Cretaceous Rivas and the Eocene Brito formations as parts of a continuous succession. The Masachapa Formation is Oligocene in age (Auer 1942, unpublished paper; Zoppis Bracci & Del Guidice 1958) and the boundary between the Brito and Masachapa formations has been interpreted as conformable (McBirney & Williams 1965). Also, the contact between the Masachapa Formation and the overlying Miocene El Fraile Formation is regarded as conformable (McBirney & Williams 1965; Kuang 1971) and the El Fraile Formation interfingers with part of the volcanic Tamarindo Formation.

The rocks of the Nicaraguan Highland are generally affected by low-grade metamorphism of zeolite facies (Levi *et al.* 1987),

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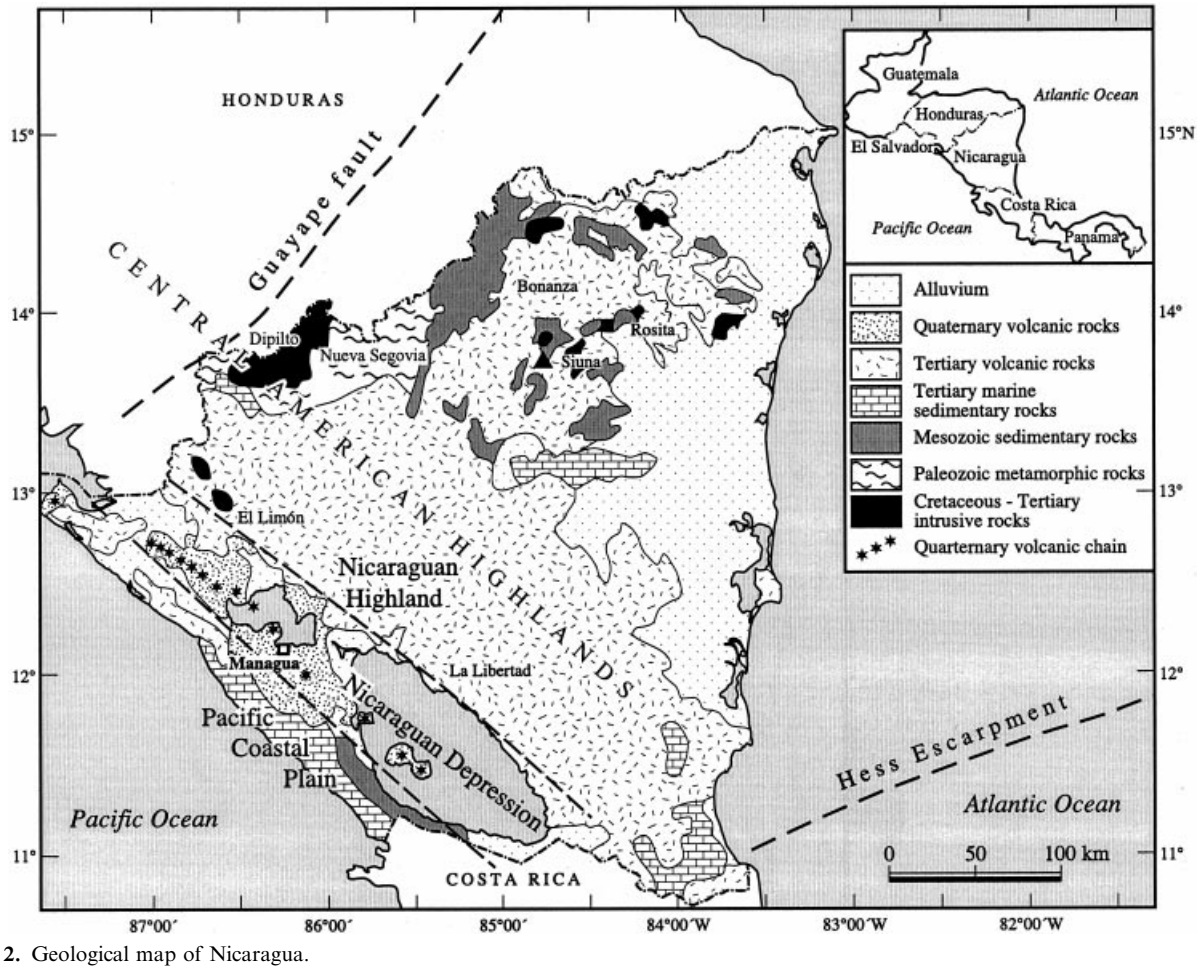


Figure 2. Geological map of Nicaragua.

which means a temperature lower than 150 °C. Also in the Pacific Coastal Plain secondary zeolite and clay minerals are found originating from alterations of volcaniclastics in the sedimentary units (Levi *et al.* 1990). The secondary minerals show a vertical zoning typical of burial diagenesis, which is a process controlled by an increase in temperature with depth. A palaeothermal gradient of about 15–20 °C km^{-1} is estimated for the Cretaceous to Oligocene diagenetic episodes and the temperature in the lower parts of the Rivas, Brito and Masachapa formations is estimated at *c.* 120 °C. Mineralogical breaks are observed between the Masachapa and Brito formations as well as between the Brito and Rivas formations. These breaks were suggested to reflect tectonic events of regional character (Levi *et al*. 1990), possibly related to changes in the polarity of subduction.

PALAEOMAGNETIC SAMPLING

Samples for the present study were collected from the Nicaraguan Highland and from the Pacific Coastal Plain. The study includes some 245 samples from 30 sites (Fig. 3), collected as block samples or with a portable drill, and oriented with sun or magnetic compasses. Generally, each sampling site represents a 10–20 m^2-scale outcrop. In the Highland, predominantly volcanic rocks from the Miocene Lower Coyol Group and from assumed Oligocene–Palaeocene rocks of the Matagalpa Group were sampled. As the tectonic control was often poor, our ambition has been to collect samples of rocks of similar age but geographically well-separated in order to detect later local tectonic movements.

To the west of the Nicaraguan Depression we collected sedimentary and intrusive rocks of the Rivas Formation, sedimentary rocks of the Brito Formation and volcanic rocks of the Tamarindo Formation. The ages of these formations are Upper Cretaceous, Eocene and Miocene–Pliocene, respectively.

MINERALOGICAL STUDIES

Petrographic and X-ray diffraction studies (Nyström & Levi 1985; Elming 1985) have been carried out on specimens of andesite, basalt and sedimentary rocks from a number of sites. Most of the Tertiary rocks of this study are only incipiently altered. As weathering is a particular problem in tropical areas, martitization, the alteration of magnetite to haematite, is a more or less common phenomenon. Martitization is here developed preferentially on the grain surfaces and in the crystals along lamella structures and in cracks. The degree of martitization varies in the samples but is better developed in rocks from some sites (e.g. sites 1, 3 and 6). Although the remanent magnetization of haematite is usually more resistant to demagnetization than magnetite, no significant relation between degree of martitization and directional variations of the characteristic magnetizations was found. Magnetite is

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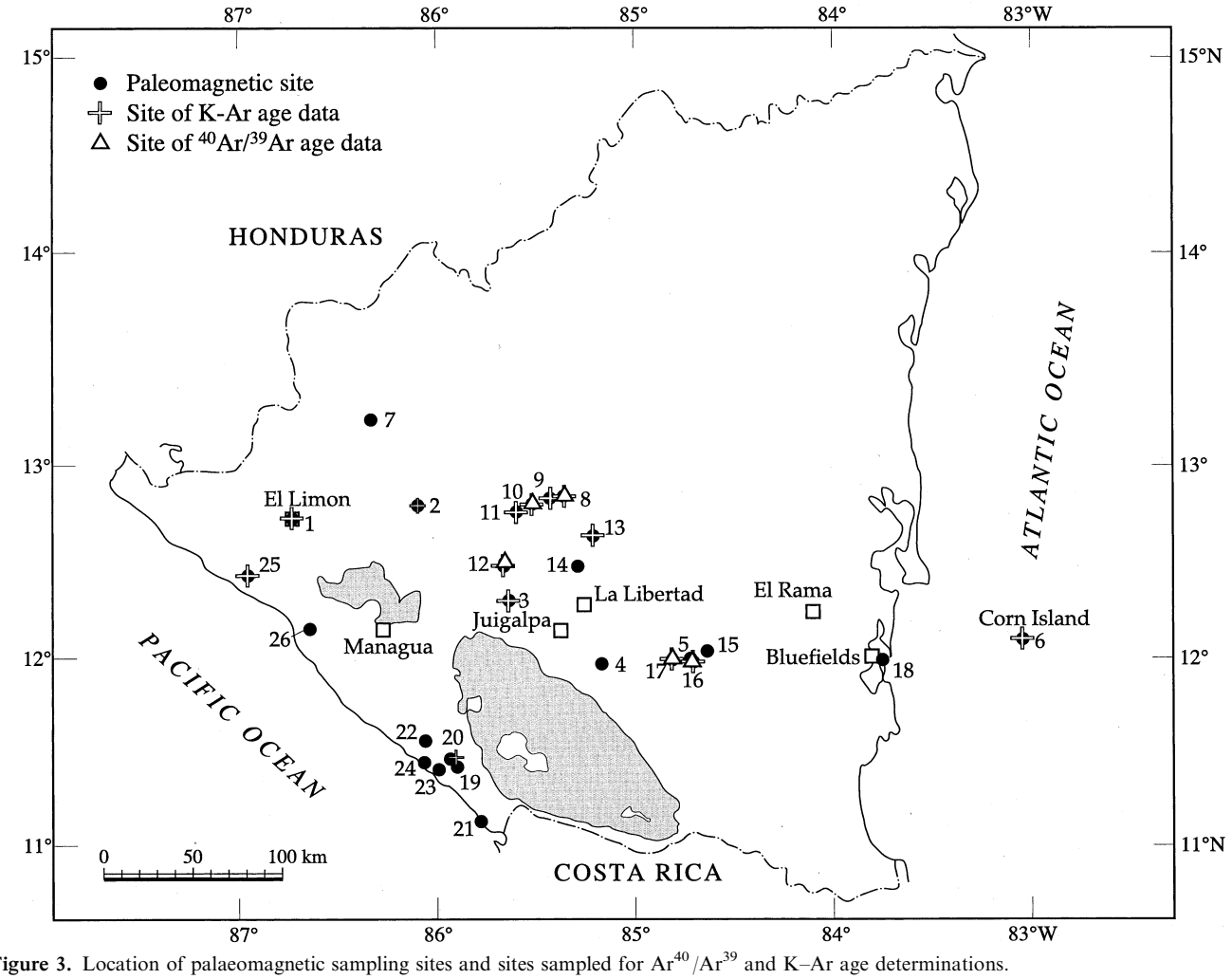


Figure 3. Location of palaeomagnetic sampling sites and sites sampled for Ar^{40}/Ar^{39} and K–Ar age determinations.

sometimes altered to maghemite (Nyström & Levi 1985); however, magnetite of primary origin is the main carrier of the remanent magnetization.

K–Ar AND $^{40}Ar/^{39}Ar$ RADIOMETRIC AGE DETERMINATIONS

In order to constrain the ages of magnetizations in this study, we have dated samples from selected palaeomagnetic sites by the K–Ar and $^{40}Ar/^{39}Ar$ methods. Previous Rb–Sr and K–Ar dating of whole-rock and mineral phases (Zoppis Bracci 1961; Barr & Escalante 1969; McBirney & Williams 1965) from both the Highland and the Pacific Coastal Plain were of poor resolution and indicate that many of the units may have been emplaced over a long duration. Dating the same samples and flows for which we have stable magnetic directions allows us to avoid assigning 'incorrect' or estimated ages to the magnetization.

The K–Ar age determinations were made by R. M. Macintyre at the Scottish Universities Research and Reactor Centre (Table 1). One–four palaeomagnetic samples from each of the 11 sites 2, 3, 6, 8, 9, 10, 11, 12, 13, 16 and 17 were analysed, including andesite and basalt from the Lower Coyol and Matagalpa groups and a diabase dyke (Table 1). Basalts from sites 12 and 14 and the basic dyke at site 10 are not identified in the field as being part of any specific group; however, based on evidence presented here, we have assigned these sites to the Matagalpa and Lower Coyol groups, respectively. From the Pacific Coastal Plain, ignimbrite and basalt were dated. One of the dated basalts was associated with palaeomagnetic site 25.

X-ray diffraction analyses and observations from thin-sections (Nyström & Levi 1985) show that most of the samples are relatively well preserved. However, these relatively unaltered rocks do contain up to 10 per cent montmorillonite and zeolite, minerals known to pose problems for K–Ar whole-rock dating. This potential for disturbance can be seen in some of the age scatter in sites where more than one age determination was made. For example, samples from sites 9 and 10 (Table 1) show significantly different ages of samples from the same flow. Although K–Ar ages from the Lower Coyol Group tend to be younger than those from the Matagalpa Group, the ages from the two groups show large ranges and overlap. Thus it is possible that either some flows have been misassigned in the field or that the K–Ar dating systematics have been disturbed in some samples due to alteration.

In order to assess the extent of alteration in the samples and the reliability of the K–Ar ages better, selected whole-rock samples were dated by the $^{40}Ar/^{39}Ar$ step-heating method. One

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Table 1. K–Ar whole-rock ages determined at the Scottish Universities Research and Reactor Centre (Macintyre in Elming 1999) and whole-rock $^{40}Ar/^{39}Ar$ ages of this study. The dated rocks are included in the palaeomagnetic study. Coordinates of the sites are given in the UTM system.

Site	Coord. (N–S/E–W)	Age (Myr)	Comments
Nicaraguan Highland			
2	1420.0/599.0	15.4 ± 0.5	Basalt, Lower Coyol Group, Sebaco, WR.
3	1361.0/648.8	25.5 ± 1.0	Andesite, Lower Coyol Group, WR.
		23.8 ± 1.3	
6	1344.0/937.0	2.8 ± 1.0	Basalt, recent, WR, poor age.
8	1420.0/678.0	27.2 ± 0.8	Basalt, Matagalpa Group, Matiguas–Rio Blanco, WR.
		27.7 ± 0.8	
		23.7 ± 0.6	
		26.0 ± 0.8	
		23.80 ± 0.14	Ar–Ar, WR.
9	1419.0/670.0	31.6 ± 1.0	Basalt, Matagalpa Group, Matiguas, WR.
		28.2 ± 1.1	
		18.2 ± 0.6	
		23.2 ± 1.2	
10	1415.5/659.5	15.9 ± 0.9	Diabase dyke.
		16.6 ± 0.9	Rio El Paso, WR.
		40.4 ± 0.9	
		19.2 ± 0.8	
		18.46 ± 0.37	Ar–Ar, WR.
11	1411.0/651.0	13.0 ± 0.4	Basalt stock, Muy Muy, WR.
		17.7 ± 0.5	
		16.9 ± 0.4	
		16.9 ± 0.4	
12	1381.9/644.0	24.5 ± 0.8	Basalt, Boaco, WR.
		17.61 ± 0.27	Ar–Ar, WR.
13	1397.3/694.5	26.6 ± 0.8	Basalt, Matagalpa Group, St Bertha, WR.
		27.4 ± 0.8	
14	1378.1/680.9	32.13 ± 0.22	Ar–Ar, WR.
16	1327.4/749.6	25.8 ± 0.8	Basalt, La Batea Formation, WR.
		22.96 ± 0.40	Ar–Ar, WR.
		22.74 ± 0.45	Ar–Ar, WR.
17	1326.2/738.3	17.8 ± 1.1	Andesite, La Batea Formation, WR.
		19.59 ± 0.25	Ar–Ar, WR.
Pacific Coastal Plain			
25	1372.9/506.7	12.3 ± 1.7	Basalt, Tamarindo Group, WR.

sample was chosen from the Lower Coyol Group (site 10), two were chosen from the La Batea Formation (sites 16 and 17) and three were chosen from other localities of the Matagalpa Group (sites 8, 12 and 14).

Samples were crushed and sieved and small chips from the 100–250 μm fraction were chosen for dating. Samples of Bern4B biotite with an assumed age of 17.25 ± 0.20 Myr were used to monitor the neutron flux. The samples were irradiated for 2 MW h in position 5c of the uranium-enriched research reactor of McMaster University in Hamilton, Ontario, Canada. The samples were step-heated from about 500 to 1600 °C. The purified argon isotopes were then analysed in a Nuclide 6-60-SGA mass spectrometer at the Geophysical Institute, University of Alaska, Fairbanks, USA. Argon purification was achieved using a liquid nitrogen cold trap SAES Zr–Al getter at 400 °C. The measured argon isotopes were corrected for system blank and mass discrimination as well as calcium, potassium and chlorine interference reactions following the procedures outlined in McDougall & Harrison (1988). Plateau ages were based on the criteria of three or more contiguous fractions accounting for more than 50 per cent gas release having ages within a 2σ (95 per cent confidence level) of the mean age.

For five samples (sites 8, 10, 12, 14 and 16), plateau ages consisting of at least three steps totalling more than 60 per cent gas release were calculated for each sample. Two samples (sites 16 and 17) contained excess argon and an isochron was fitted to the data. For these samples an initial $^{40}Ar/^{36}Ar$ value of *c.* 310 was found. A summary of all the $^{40}Ar/^{39}Ar$ results is given in Fig. 4, with all ages quoted to the $\pm 1\sigma$ level and calculated using the standards of Steiger & Jäger (1977). Detailed geochronological data are available upon request from the authors.

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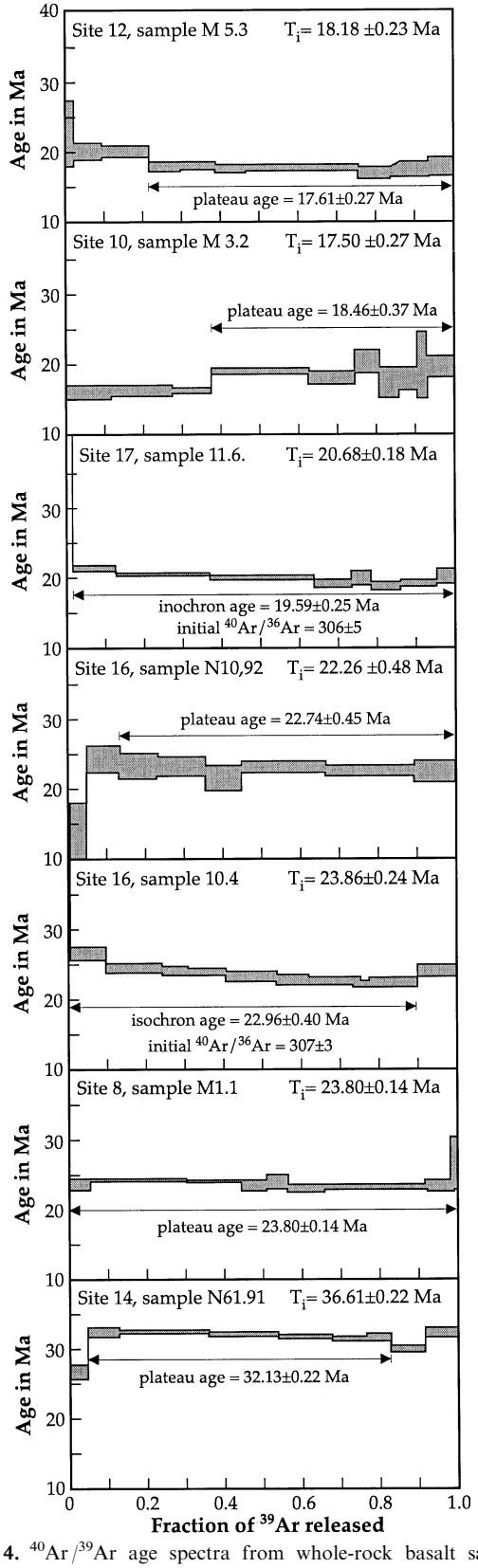


Figure 4. $^{40}Ar/^{39}Ar$ age spectra from whole-rock basalt samples. Samples are designated by palaeomagnetic sampling sites and sample number. Spectra are arranged with the youngest age at the top and the oldest at the bottom. The integrated (total gas) age is denoted by T_i. The plateau or isochron ages have been calculated from selected fractions designated by arrows and these ages are used in the discussion. See text for analytical procedures.

Pacific Coastal Plain

One K–Ar age from a palaeomagnetic site of the Tamarindo Group has an age of 12.3±1.7 Myr for a basalt (site 25; Table 1). Three additional samples from other locations of the Tamarindo Group not associated with palaeomagnetic sites have ages of 18.4±4.9 Myr (rhyolite ignimbrite), 15.4±0.5 Myr (acid ignimbrite) and 13.7±0.8 Myr (basalt) (Elming 1999). The weighted average age for these samples is 14.8±0.5 Myr.

Nicaraguan Highland

Eight of the 12 K–Ar age analyses of Lower Coyol rocks are from basalt flows and cluster about a single mode with a weighted average age of 16.9±0.4 Myr. Of the remaining four analyses, two (a 13 Myr age from site 11 and a 40 Myr age from site 10) are not consistent with other more reliable samples within the site and probably reflect argon loss or gain due to alteration. Two samples from site 3 are from andesite rather than basalt and they have an average age of 24.9±0.8 Myr. These ages are significantly different from the average of the basalt ages. It is not known if this difference reflects two stages of volcanism in the Coyol Group or if the andesite samples have incorporated excess argon and hence the ages are anomalous.

The spectrum of a single $^{40}Ar/^{39}Ar$ measurement from site 10 (Fig. 4) shows evidence of argon loss and the sample has an integrated age of 17.5±0.3 Myr. The integrated age of this sample is similar to the average K–Ar age, which implies that many of the Coyol samples may either have experienced some argon loss or have been emplaced over a period of *c.* 2 Myr. The plateau age of this sample is 18.5±0.4 Myr and we interpret this age to reflect more closely the true age of this rock. In the absence of further geochronological information to constrain the age of the Lower Coyol Group, we assign it an age of *c.* 18 Myr.

The 13 K–Ar ages from the Matagalpa Group range from about 18 to about 31 Myr (Table 1). 10 of the 13 ages form a poorly constrained mode with an average age of 25.9±1.1 Myr. It is probable that the samples of the Matagalpa Group reflect multiple events (e.g. Ehrenborg 1996; Hodgson 1999) and an average age of all sites is not geologically significant.

Within the Matagalpa Group, the La Batea Formation appears to be the youngest unit. Samples from sites 16 (basalt) and 17 (andesite) of this formation have K–Ar ages of 25.8 and 17.8 Myr. Three samples were dated by the $^{40}Ar/^{39}Ar$ method. Two of the samples show evidence of excess argon, probably from a later alteration event. The average age for the basalt unit (site 16) is about 23 Myr, while the age of the andesite (site 17) is 19.6 Myr, significantly younger.

Based on the palaeomagnetic evidence (presented below), the remaining Matagalpa samples can be divided into two groups, sites 8, 9 and 12, and sites 13 and 14. 11 K–Ar ages from the first group show a 'bimodal' distribution with two ages of *c.* 18 Myr and nine ages scattering between 23 and 32 Myr, with a mean K–Ar age of 26.5±0.9 Myr. An $^{40}Ar/^{39}Ar$ age of 23.80±0.14 Mr is seen for a single sample from site 8, consistent with the group of older ages. A sample from site 12 has an $^{40}Ar/^{39}Ar$ age of 17.61±0.24 Myr, consistent with the younger age group. These young ages are similar to ages seen in flows mapped as Lower Coyol, and it is possible that the younger ages seen at these sites represent a second pulse of magmatic activity.

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Samples from sites 13 and 14 form the second palaeomagnetic group. The mean K–Ar age from site 13 (two samples) is 27 Myr, while an $^{40}Ar/^{39}Ar$ plateau age of 32.1 ± 0.2 Myr is seen from site 14 (Fig. 4). Although it is impossible to assess the amount of argon loss from samples of site 13, it is possible that these samples are significantly older than those from other Matagalpa sites and have an age of *c.* 30 Myr.

Age summary

Based on our new geochronological evidence, most of the samples dated by the $^{40}Ar/^{39}Ar$ method show either significant argon loss or significant argon gain. If these signs of alteration are prevalent throughout the sample suite, then interpretation of the K–Ar ages is difficult. Nevertheless, we can infer that the age of the Tamarindo Group of the Pacific Coastal Plain is *c.* 15 Myr, samples correlative with the Lower Coyol Group are *c.* 18 Myr, the La Batea member of the Matagalpa Formation is *c.* 20 Myr, Matagalpa sites 8, 9 and 12 are *c.* 24 Myr and sites 13 and 14 are *c.* 30 Myr.

PALAEOMAGNETIC RESULTS

The specimens were demagnetized by conventional alternating field (AF) and thermal demagnetization techniques and remanent magnetizations measured with spinner and cryogenic magnetometers. The different components of remanence were defined by principal component analyses (Kirschvink 1980), and vectors defined with mean angular deviations larger than 7° have been rejected. The site mean directions and formation means were calculated using Fisher (1953) statistics.

It was possible to isolate well-defined high-stability magnetizations in 26 of the 30 sites that were sampled. The results from some sites were discarded due to very poor within-site groupings of directions. Usually there were no more than two components of remanent magnetization (Fig. 5), one of which was of low coercivity with a direction similar to that of the present Earth field (PEF; $D = 0°$, $I = 43°$).

The Nicaraguan Highland

Rocks assumed to be of the Coyol Group or of similar age have been sampled at seven sites (2, 3, 4, 7, 10, 11 and 18; Fig. 3). The coercivity of the remanence varies but the soft portion is generally erased at *c.* 20 mT (Figs 5a and b). Although the number of samples in each site is often small, the directions of the highest-stability component, referred to as the characteristic magnetization (Butler 1992), usually form well-defined means (Table 2; Fig. 6a) and they are of predominantly reversed polarity when compared to the PEF. It seems possible to distinguish two groups of directions. One (sites 2, 3, 4 and 7) is characterized by negative inclinations and the other (sites 10, 11 and 18) by shallow and positive inclinations. However, the gradual change from positive to negative directions may reflect a trend of changing magnetization ages, rather than two distinctly different times of magnetization. Therefore, these sites are treated together here (mean of sites: $D = 180.2°$, $I = -2.8°$, $\alpha_{95} = 11.9°$; Table 3), representing one magnetization age. On the basis of the age data presented here, the age of magnetization is estimated at *c.* 18 Myr.

The rest of the rocks sampled in the Nicaraguan Highland are assumed to be of Matagalpa age, i.e. older than *c.* 20 Myr. The characteristic magnetizations are of dual polarity; however, reversed polarities dominate. The inclinations are generally rather shallow, i.e. less than $c. \pm 30°$, and on changing the magnetizations to reversed polarity the declinations vary from *c.* 90° to 150°E.

The basalt and andesite of the La Batea Formation (sites 5, 15, 16 and 17) are all sampled from the central southern part of the Highland. The characteristic magnetization is generally of high coercivity and the blocking temperatures indicate the carrier to be magnetite–titanomagnetite (Fig. 5d). At site 17 the remanence is of lower coercivity, but the direction is still stable up to 65 mT. Only at this site is the magnetization of normal polarity. When changing the polarity of the site mean directions to reversed, the declinations of the site means are quite similar, while the inclinations are different. The fairly big differences in inclinations may indicate local tectonic disturbances. As the inclinations are shallow, declinations would be little affected by tilting; however, we have no evidence of tilting from field observations. The site mean directions form a mean for the La Batea Formation ($D = 148.8°$, $I = 5.9°$, $\alpha_{95} = 16.1°$; Table 3) significantly different from that of the Coyol Group (Fig. 6c). The whole-rock K–Ar and $^{40}Ar/^{39}Ar$ data suggest a magnetization age for this formation of *c.* 20 Myr.

Matagalpa rocks were also sampled in the central parts of the Highland (sites 8, 9, 12, 13 and 14; Fig. 3). These rocks yield directions of magnetization (Fig. 6a) of reversed and dual polarities, significantly different from those of the La Batea Formation. The declinations vary from *c.* 90° to 140°E and the inclinations are predominantly negative and lower than *c.* 30°. It is possible to distinguish two groups of directions with significantly different declinations of *c.* 130°E (sites 8, 9 and 12) and 90°E (sites 13 and 14), respectively. The rocks of the two groups are sampled from geographically well-separated areas (Fig. 3).

The coercivity of remanence of the first group varies but the directions are fairly stable after being demagnetized in 10 mT AF. The rock at site 8 carries characteristic magnetizations of dual polarity, which may indicate that the rock at this site has not been remagnetized. As for the La Batea Formation the mean directions of the sites (Table 2) show that the inclinations are different while the declinations are quite similar. The direction of magnetization at one of the sites (site 12) of this grand mean is also anomalous with respect to its $^{40}Ar/^{39}Ar$ age. The mean direction for this group of sites is also very poorly defined ($\alpha_{95} = 40.2°$; Table 3) and cannot be used to represent any specific age of magnetization.

The coercivity of the characteristic magnetization of the second group is usually high and in some samples the magnetization is of dual polarity (Fig. 5c), possibly indicating a primary origin of magnetization. The two site means are not significantly different from each other and the directions probably belong to a single population. The mean of the sites ($D = 93.0°$, $I = -22.0°$) is also very close to the mean of the samples ($D = 92.9°$, $I = -17.9°$, $\alpha_{95} = 4.0°$; Table 3). The age data suggest magnetization ages for this second group of 30 Myr.

Basalt of only a few million years in age (site 6) has been sampled off the Caribbean coast of Nicaragua (Fig. 3). This rock carries a stable remanent magnetization with a direction ($D = 346°$, $I = 22°$) similar to that of the theoretical dipole field (Fig. 6a; Table 2).

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Table 2. Palaeomagnetic site mean directions and calculated pole positions.

Site	n	D/I	α_{95}	k	P	Plat/Plon	A_{95}	Comments
Nicaraguan Highland								
1	4	203/−28	8.4	121	R	68/355	6.8	Dacite
2	5	186/−14	4.4	301	R	82/49	3.2	L. Coyol, basalt
3	5	179/−7	14.9	27	C	81/103	10.6	L. Coyol, andesite
4	3	172/−2	14.3	75	R	76/135	10.1	Basalt, L. Coyol?
5	3	151/18	14.9	69	R	57/158	10.7	La Batea, basalt
6	5	346/22	9.9	60	N	77/186	7.7	Recent, basalt
7	5	179/−27	7.1	117	R	89/237	5.7	UCoyol?, dacite
8	9	130/−23	10.7	24	C	41/189	8.2	Matagalpa, basalt
9	6	125/13	7.7	76	R	32/168	5.6	Matagalpa, basalt
10	15	187/8	2.9	181	R	72/72	2.1	Diabase, Muy-Muy
11	5	181/3	6.4	145	R	76/91	4.5	Stock, Muy-Muy
12	6	137/−34	5.7	141	R	48/198	4.9	Basalt, Boaco
13	8	93/−29	13.6	18	R	9/199	11.0	Matagalpa, basalt
14	26	93/−15	2.9	97	R	4/192	2.1	Basalt, La Kalamidad
15	6	147/−9	6.4	109	R	56/176	4.6	Bat. Fm, basalt
16	12	144/17	4.3	103	R	48/157	3.2	Bat. Fm., basalt
17	7	334/2	5.6	116	N	62/164	4.0	Bat. Fm., andesite
18	9	178/19	2.4	480	R	69/102	1.8	Basalt, El Bluff
Pacific Coastal Plain								
19	5	356/34	13.1	35	N	82/244	11.3	Rivas Fm., dyke
20	7	184/−23	7.5	65	C	86/355	5.3	Rivas Fm., dyke
21	5	192/−17	12.3	40	C	75/7	9.3	Brito Fm.
22	4	15/7	24.4	15	C	73/30	17.4	Brito Fm.
23	10	176/6	13.4	14	R	75/111	9.5	Brito Fm.
24a	18	10/28	4.7	55	N	80/343	3.8	Brito Fm.
24b	11	9/4	5.2	77	N	77/49	3.7	High Hc.
25	19	186/−23	3.2	109	R	85/4	2.5	Tam. Fm., basalt
26	11	356/19	5.7	65	N	85/151	4.3	Tam. Fm., andesite
Previous work in the Pacific Coastal Plain of Nicaragua								
G1	18	353/32	4.1	72	N	81/227	3.5	Rivas Fm.
G2	11	2/22	2.8	262	N	88/262	2.2	Rivas Fm.
G3	20	176/−16	4.0	57	R	85/143	2.9	Brito Fm.

Site = name of site; n = number of samples; D = mean declination, I = mean inclination; α_{95} = radius of 95 per cent confidence circle; k = Fisher precision parameter; P = polarity—reversed (R), normal (N), combined (C); Plat, Plon = latitude and longitude of calculated palaeomagnetic pole positions; A_{95} = radius of the 95 per cent confidence circle of the mean pole; Comments = rock group, rock type or location. The data from sites G1–G3 are from Gose (1983).

The virtual geomagnetic pole (VGP) positions (Fig. 7) calculated from the site means and the grand means of the Highland show the same trend of distribution as poles calculated from Cretaceous rocks in Honduras (Gose 1985; Gose & Finch 1992). The centres of the polar arcs coincide with the sampling region, which means that the distribution of poles may have its origin in local structural rotations as well as plate tectonic rotations (MacDonald 1978, 1990).

The Pacific Coastal Plain

From the Pacific Coastal Plains, basalt (site 25) and andesite (site 26) of the Tamarindo Group have been sampled. The site mean directions are of both reversed ($D = 186^\circ$, $I = -23^\circ$, site 25; Fig. 6b; Table 3) and normal ($D = 356^\circ$, $I = 19^\circ$, site 26) polarities and on inverting the polarity of one they are not significantly different. After removing a soft component of remanence with a direction similar to that of the PEF, the magnetization of the basalt is of high coercivity and the directions are stable up to 100 mT AF (Fig. 5e). The portion of soft remanence is much smaller in the andesite and a stable magnetization is isolated after treatment in *c.* 25 mT AF. The mean direction of the two sites ($D = 0.9^\circ$, $I = 21.1^\circ$) is not significantly different from the mean of the samples ($D = 2.0^\circ$, $I = 21.6^\circ$, $\alpha_{95} = 3.2^\circ$; Table 3). The Tamarindo Group is regarded as equivalent to the Coyol Group of the Highland and an age of *c.* 15 Myr is estimated for the magnetization on the basis of the age data presented here.

Sedimentary rocks of the Brito Formation have been collected at five sites and at four of them (21, 22, 23 and 24; Fig. 3) a characteristic magnetization could be defined. Bedding correction has been performed and the mean directions presented here all refer to directions after structural correction. At sites 23 and 24, samples were taken from stratigraphic intervals of 7 and 5.5 m, respectively. The sandstone and shale at site 23 often carry a large portion of low-blocking-temperature (T_b) components erased at temperatures below 200 °C and a characteristic magnetization is isolated at temperatures between 300 and 450 °C. This magnetization, which is of reversed polarity, was defined during thermal demagnetization in 10 of the 24 samples collected at this site. No stable magnetization was isolated during AF demagnetization. The bedding dips *c.* 13°SW and the corrected mean direction ($D = 176^\circ$, $I = 6^\circ$; Table 2) has a shallow inclination. No significant directional difference can

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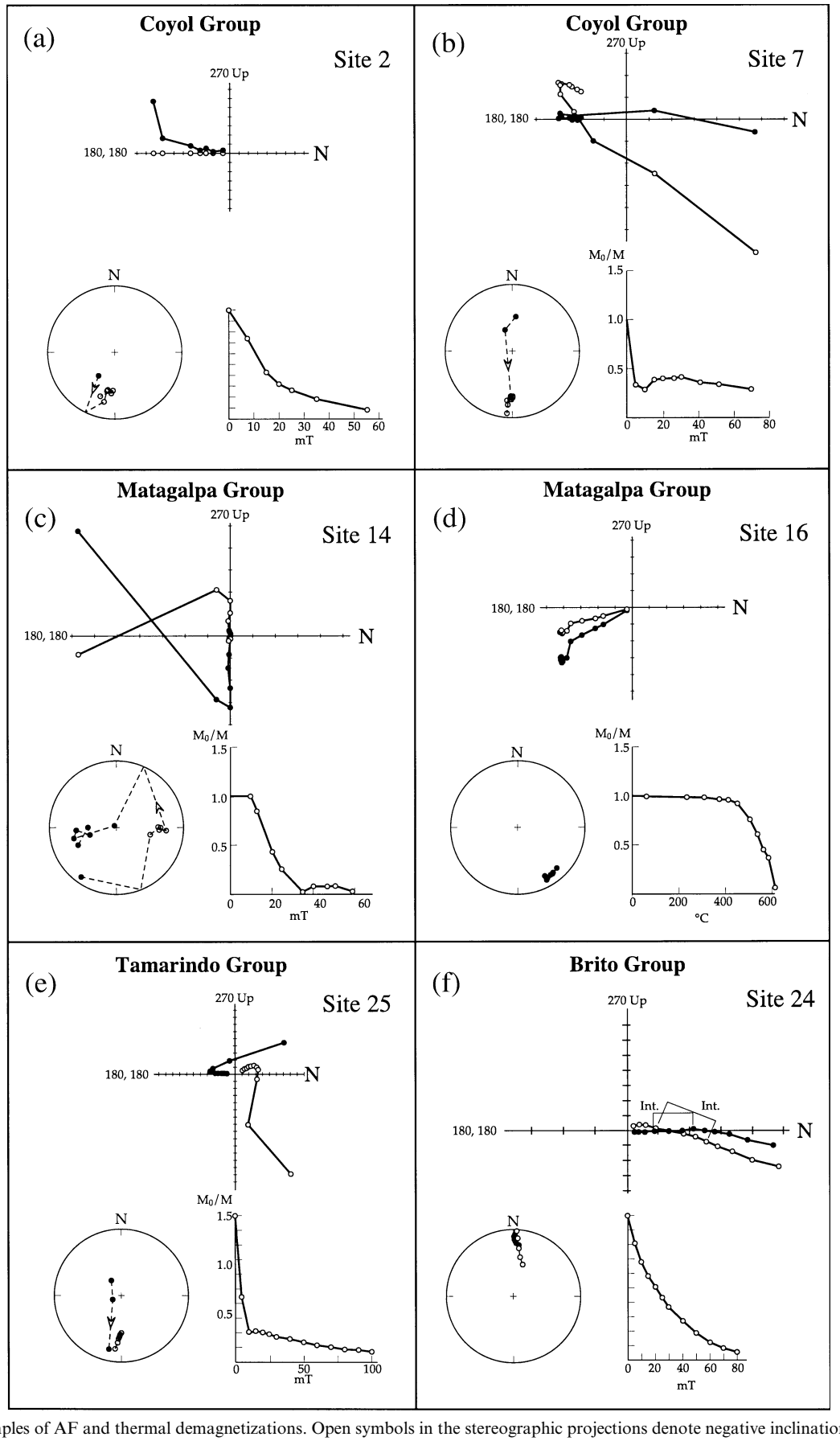


Figure 5. Examples of AF and thermal demagnetizations. Open symbols in the stereographic projections denote negative inclinations. In the vector plots open symbols denote inclinations, while filled symbols denote declinations.

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Table 3. Mean directions of sites and calculated grand mean poles for the different rock groups/formations.

Sites	*N*/*n*	*D*	*I*	α_{95}	*k*	P	Plat./Plon.	A_{95}	Age(Myr)
Nicaraguan Highland									
COYOL GROUP									
2,3,4,7,10,11,18	7*/47	180.2	−2.8	11.9	27	C	78.9/93.5	8.4	18
MATAGALPA GROUP									
La Batea Fm.									
5,15,16,17	4*/28	148.8	5.9	16.1	33.3	C	55.6/161.2	11.5	20
Matagalpa Gr.									
8,9,12	3*/21	130.3	−15.0	40.2	10.5	C	40.9/183.8	29.6	24
Matagalpa Gr.									
13,14	2*/34	93.0	−22.0			R	5.8/195.0		30
	2/34*	92.9	−17.9	4.0	39.8	R	4.8/192.9	3.0	
Pacific Coastal Plain									
TAMARINDO FORMATION									
25,26	2*/30	0.9	21.1			C			15
	2/30*	2.0	21.6	3.2	67.5	C	87.7/34.3	2.5	
BRITO FORMATION									
(B_1)	4*/30	8.1	5.8	14.4	41.5	C	80.9/67.0	11.5	Eocene
21,22,23,24hHc		8.2	2.0	23.1	16.7	C	76.9/55.0	16.4	
(B_2)	5*/50	5.8	7.9	12.3	39.5	C	80.8/55.3	8.8	
21,22,23,24hHc,G3		6.3	2.5	17.1	21.1	C	78.1/62.1	12.1	
(B_3)	4*/46	3.4	8.0	15.6	35.8	C	82.1/68.8	11.1	
21,23,24hHc,G3		4.4	−0.6	21.3	19.6	C	77.5/73.4	15.1	
RIVAS FORMATION									
19,20	2*/12	0.2	28.6				86.1/277.1		*c.*50
	2/12*	0.9	27.7	7.0	39.6	C	86.6/288.9	5.6	
19,20	4*/41	358.9	27.8	8.7	113.2	C	86.5/256.7	7.0	UC–EE
G1,G2		1.0	23.4	7.7	142.4	C	88.7/324.5	6.0	

N = number of site means, *n* = number of samples; Age = age of magnetization estimated on the basis of the K–Ar and $^{40}Ar/^{39}Ar$ ages presented here. For the formation means of the Pacific Coastal Plain, the first entry is corrected for bedding and the second entry is without structural correction. For other symbols see footnote to Table 2.

be detected when comparing the results from the different stratigraphic levels.

At site 24 two or three different components of magnetization can be identified during AF demagnetization. The coercivity of remanence varies in the sandstone samples and is lower in the intermediate part of the stratigraphic sequence. In the lower parts, the coercivity spectrum, expressed by the medium destructive field (MDF) (Dunlop & Özdemir 1997), varies between 10 and 23 mT and three components have been isolated (Fig. 5f). A soft magnetization with a direction not far from the theoretical dipole direction is erased in 15 mT AF. A second component is isolated in 20–50 mT AF and 350–450 °C thermal treatment; however, the mean direction of this magnetization (site 24a, $D = 10°$, $I = 28°$; Table 2) is not significantly different from the mean direction of the soft component. A third direction is isolated after demagnetization in 40–100 mT AF. The direction of this magnetization ($D = 9°$, $I = 4°$; Table 2) is significantly different from those of lower coercivities. For some samples the trajectories in the vector diagrams do not pass through the origin (Fig. 5f), which indicates the existence of an undefined magnetization. At increasing AF demagnetization levels the directions in these samples change along great-circle paths. The remagnetization circles were analysed using the method of Halls (1976) and the great circles intersect at a declination and inclination of $D = 191.1°$, $I = -12.5°$, which suggests the existence of a reversed-polarity high-coercivity magnetization.

The characteristic magnetization carried by the sandstone at site 21 ($D = 192°$, $I = -17°$; Table 2) is isolated during thermal demagnetization at temperatures above 500 °C. A low-temperature component with a direction similar to that of the PEF was erased at temperatures below 400 °C.

The site mean direction for site 22 is poorly defined ($D = 15°$, $I = 7°$, $\alpha_{95} = 24.4°$; Table 2). The characteristic magnetization is isolated at temperatures of 450 °C and higher. The directions are of mixed polarity and imperfect cleaning and isolation of the different components may be the reason for the poorly defined mean direction.

After bedding correction, the directions for sites 21, 22, 23 and the high-coercivity component at site 24 were used to calculate a mean for the Brito Formation (B_1: $D = 8.1°$, $I = 5.8°$; Table 3) with a Fisher (1953) precision parameter (k) of 41.5. If the intermediate-coercivity component of site 24 is used for the mean, the precision parameter is lower ($k = 23.9$). This probably indicates that the high-coercivity component is primary and that the intermediate-coercivity component is of secondary origin. Gose (1983) reported palaeomagnetic results from the Brito and Rivas formations in the southwestern part of Nicaragua. If his result from the Brito Formation (site G3, Table 2) is included in the mean, the new formation mean (B_2: $D = 5.8°$, $I = 7.9°$; Table 3) is not significantly different from the former mean (B_1) and the precision ($k = 39.5$) is similar. If the poorly defined direction for site 22 is excluded, the mean direction is similar (B_1: $D = 3.4°$, $I = 8.0°$; Table 3), but, the

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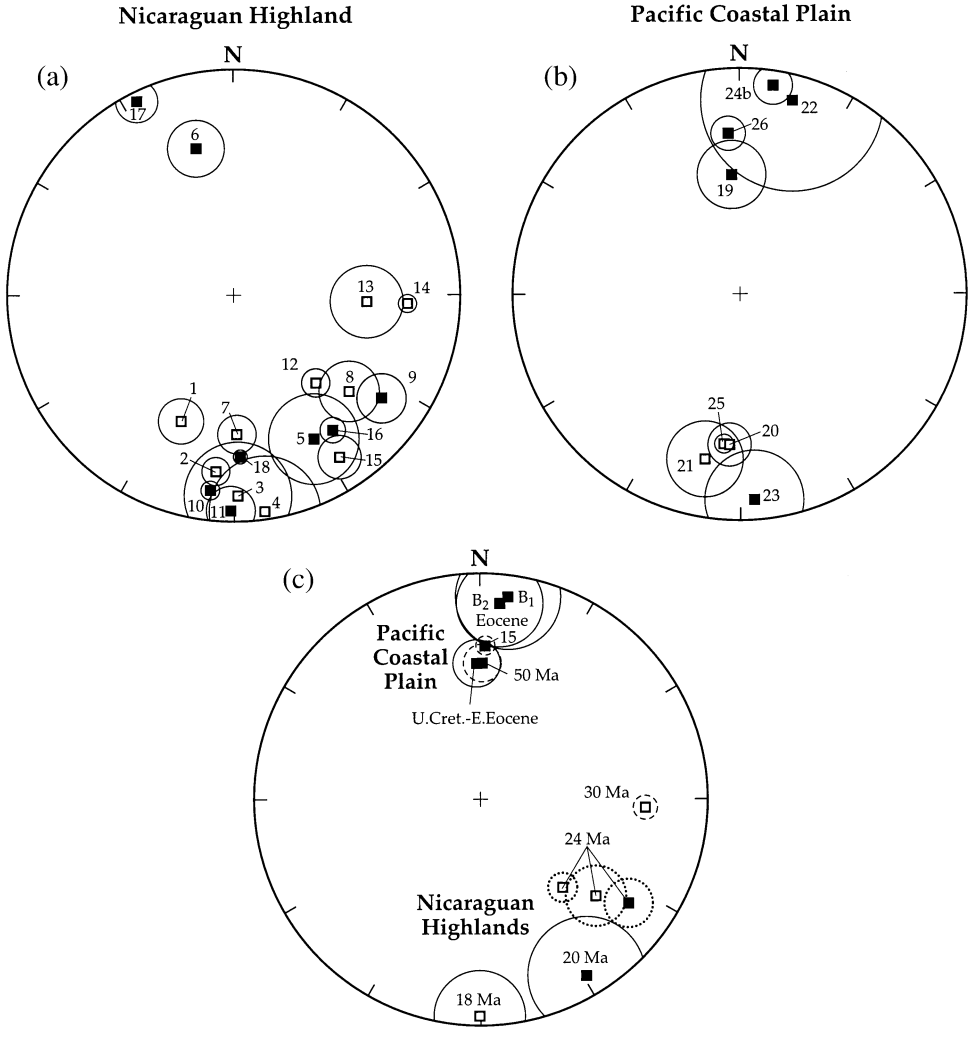


Figure 6. Site mean directions and corresponding circles of 95 per cent confidence for rocks sampled in (a) the Nicaraguan Highland and (b) the Pacific Coastal Plain. (c) Mean directions for the rock groups/formations together with age of magnetization. Circles of confidence marked by dashed lines refer to sample mean directions for two sites and those marked by dots refer to individual site means. Note the gradual change in declination and significantly different rock group mean directions with increasing age of rocks from the Nicaraguan Highland.

precision is somewhat lower ($k=35.8$). Even though the precision parameter increases after bedding correction (Table 3), it is not possible to demonstrate positive results of fold tests at the 95 per cent level for any of the mean directions, as can be seen from the ratio of the precision parameters (Fisher 1953) after (k_2) and before (k_1) correction (B_1: $n=4$, $k_2/k_1=2.49$; B_2: $n=5$, $k_2/k_1=1.87$; B_3: $n=4$, $k_2/k_1=1.83$; McElhinny 1964). A possible reason for this may be the small dip of the layers.

Two basic rocks (sites 19 and 20) intruding the Rivas Formation were sampled. At site 19 the basic dyke carries a remanence of rather low coercivity (MDF = 8 mT); however, a characteristic magnetization can be defined after AF cleaning in fields higher that 15 mT. The mean direction ($D=356°$, $I=34°$; Fig. 6b; Table 2) is of normal polarity and not far from that of the PEF. In one sample a high-coercivity component of reversed polarity is isolated.

The coercivity of the characteristic magnetization in the basic sill at site 20 is high (MDF = 35 mT) and the direction is stable after erasing a soft component with a direction close to that of the PEF. The direction of the characteristic magnetization is of reversed polarity ($D=184°$, $I=-23°$; Table 2). The fact that there are mixed but predominantly normal polarities at site 19 and reversed polarities at site 20 suggests that the magnetizations are not a result of remagnetization. The mean of the sites ($D=0.2°$, $I=28.6°$; Table 3) is not significantly different from the mean of the samples ($D=0.9°$, $I=27.7°$). The basic sill has been radiometrically dated at 50 Myr (K–Ar; Barr & Escalante 1969). Gose (1983) collected sandstone samples at two sites from the Rivas Formation. After structural correction the site mean directions (G1: $D=353°$, $I=32°$; G2: $D=2°$, $I=22°$; Table 2) are very close to the site means of this study. The K–Ar age of the basic sill is significantly younger than the assigned Upper Cretaceous age of the Rivas Formation. This means either that there was no apparent polar wander between the Upper Cretaceous and the Lower Eocene, or that the K–Ar age is incorrect. As the site mean directions are close to the direction of the theoretical dipole field another possibility is that the rocks have been remagnetized (for discussion see next section). If the magnetizations are primary, the formation mean direction ($D=358.9°$, $I=27.8°$; Table 3) calculated from the four site means represents a magnetization of Upper Cretaceous–Lower Eocene age.

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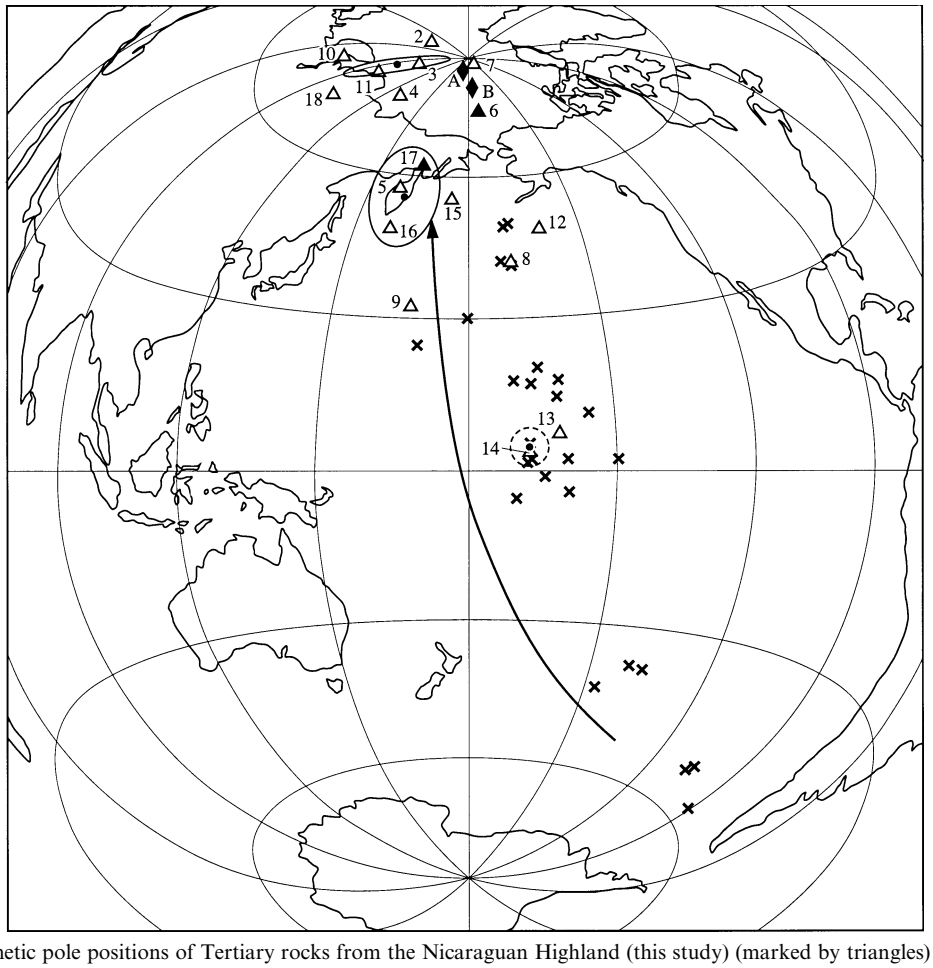


Figure 7. Palaeomagnetic pole positions of Tertiary rocks from the Nicaraguan Highland (this study) (marked by triangles) and Cretaceous rocks from Honduras (Gose 1985; Gose & Finch 1992) (marked by crosses). Open symbols denote poles of reversed polarity and the filled symbols A and B denote the 30 and 60 Myr reference poles for North America. Grand mean poles are marked by dots and shown with their circles of confidence (A_{95}). The arrow indicates the direction of decreasing age of the poles.

The palaeomagnetic poles calculated from the grand mean poles of the Tamarindo, Brito and Rivas formations all cluster around the geographical pole.

ORIGIN OF MAGNETIZATION

The pole positions calculated from the characteristic magnetizations of the rocks in the Highland and the Pacific Coastal Plain sometimes fall close to the geographical pole, which may imply that the rocks have been remagnetized.

Mineralogical studies of rocks from the Highland indicate only low-temperature ($\leq$120 °C) burial metamorphism and predominantly unaltered magnetite as the magnetic mineral. Also, the palaeomagnetic study shows that magnetite is the dominant carrier of remanence. Usually there are only two components of magnetization even though the coercivity and blocking-temperature spectra are often rather wide (e.g. Fig. 5d). The soft components often have directions similar to that of the PEF, indicating this magnetization to be a viscous remanence. In the Highland there are remanent magnetizations of both polarities and at some sites there are magnetizations of mixed polarity. Although reversals are no guarantee of a primary magnetization, they support an interpretation of a primary origin of the characteristic magnetizations (Van der Voo 1990). The burial metamorphism seems not to have affected the primary, high-blocking-temperature magnetization of the rocks, although it is probably the reason for the disturbed K–Ar system resulting in the wide scatter of the measured K–Ar ages.

Also in the Pacific Coastal Plain, the rocks have been affected by burial metamorphism, but the temperatures did not exceed *c.* 120 °C. Due to probably slow erosion and uplift this relatively low temperature may have affected grains with laboratory unblocking temperatures of some 250 °C (Pullaiah *et al*. 1975). Both Rivas and Brito sediments are volcanogenic, which means that magnetite may be expected as a carrier of a detrital remanence. This is also supported by the unblocking temperatures of the Brito sandstone and shale. A large portion of the remanence is erased at temperatures below 250 °C and the characteristic magnetization is defined in the temperature range 300–580 °C. This means that the characteristic magnetization is probably a detrital magnetization not affected by burial metamorphism. The Brito sedimentary rocks carry magnetizations of dual polarity and Gose (1983) reported a magnetostratigraphic zonation of normal and reversed polarities in the Brito Formation, which most probably rules out the possibility of remagnetization. When structural correction is performed the precision of the mean direction for the Brito Formation increases, as indicated by the ratio of the precision parameters after (k_2) and before (k_1) correction (k_2/k_1 = 1.83–2.49). Even if this increase is not sufficient to show that

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the magnetization pre-dates folding at the 95 per cent confidence level, it adds credibility to the data. Gose (1983) argues for a primary origin of the detrital magnetization in the Rivas sandstone, on the basis of palaeomagnetic data, and a volcanogenic origin for the carrier of remanence. Also, the basic rocks intruding the Rivas Formation carry magnetizations of dual polarity, which supports the interpretation of a primary origin of the magnetization.

For the magnetization of the volcanic rocks, secular variations of the magnetic field must be considered. These variations are probably averaged out when the grand means are calculated from four site means or more. Three of the grand means are based on only two site means. In the Pacific Coastal Plain the two basic rocks (sites 19 and 20) intruding the Rivas Formation and the andesite and the basalt from the Tamarindo Formation (sites 25 and 26) have magnetizations of dual polarity. These magnetizations are very close to being antipolar, which suggests only minor deviations from a dipole field. The third grand mean calculated from only two site means is for the *c.* 30 Myr old rocks from the Nicaraguan Highland. At one of these sites (site 14) the rock carries a magnetization of mixed polarity, suggesting that it was not extruded at the same time as the rock at the other site. Nonetheless, the directions are not significantly different, which possibly indicates secular variations only within the error bars of mean direction.

INTERPRETATION AND DISCUSSION

The interpretation of the palaeomagnetic data of this study is based on the premise that the magnetizations are primary.

As the tectonic control of the rocks in the Highland is poor, the directions referred to here are *in situ* directions. This may also be the reason for the sometimes quite large differences between inclinations of the site means, which then form poorly defined grand mean poles. The 24 Myr grand mean pole (Table 3) is discarded due to the very poor precision in the mean direction. The tectonic reconstructions should therefore here be regarded as tentative.

In the Nicaraguan Highland, data from volcanic rocks older than *c.* 18 Myr are sparse and together with the lack in structural control, this implies that local block rotations cannot be excluded as a source of the rotation indicated by the palaeomagnetic data. A large block delineated by well-defined fracture zones has been defined in the northeastern part of Nicaragua (Leyton 1994), but no signs of rotation were observed. Normal and to some extent strike-slip faulting has been reported from geological mapping of the Highland (Ehrenborg 1996), but no evidence of structural rotations. Rocks of similar ages and directions of magnetization have been sampled at geographically well-separated sites (e.g. the La Batea Formation). This means that the rotations, indicated by the gradual change in declinations and pole positions (Figs 6 and 7) with increasing age of the rocks, should not originate from local blocks. We therefore suggest that the rotation indicated from the data in the Nicaraguan Highland is a plate tectonic rotation. From the comparison of VGPs with those of North America (Fig. 7) it is evident that this rotation, which took place between *c.* 30 and 18 Ma, is counterclockwise and *c.* 90° (Fig. 8). The rate of rotation, 7.5°Myr^{-1}, is high and may be an artefact due to the problem of determining the exact magnetization ages.

Gravity data (Ubieta 1995; Elming & Rasmussen 1997) indicate that the Mesozoic–Palaeozoic crust of the Chortis Block continues southwards beneath the Tertiary cover of volcanic rocks to the central parts of the Highland and westwards to the Nicaraguan Depression. Pb isotope data (Sundblad *et al.*

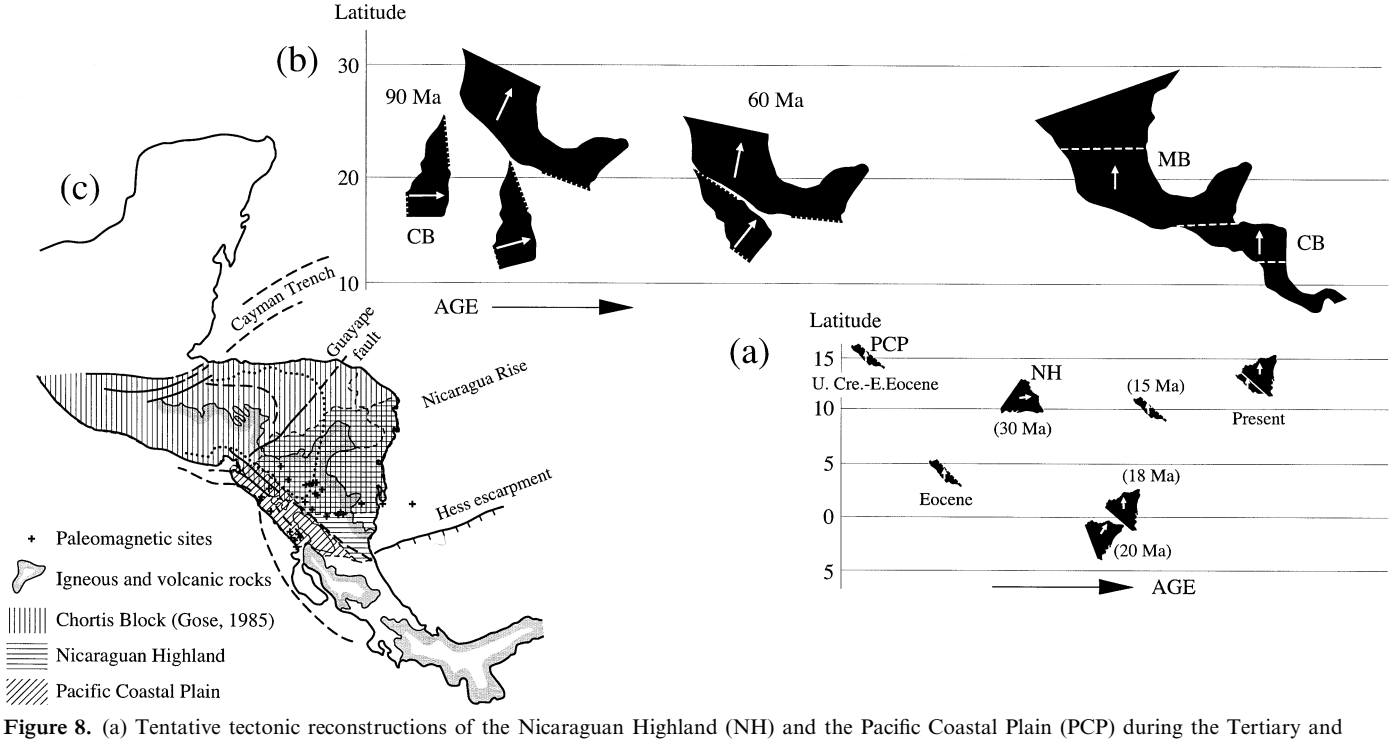


Figure 8. (a) Tentative tectonic reconstructions of the Nicaraguan Highland (NH) and the Pacific Coastal Plain (PCP) during the Tertiary and (b) palaeopositions of the Maya (MB) and Chortis (CB) blocks during the Late Cretaceous modified from Gose (1985). The areas related to the Chortis Block, the Nicaraguan Highland and the Pacific Coastal Plain are defined by vertical, horizontal and inclined lines, respectively, in (c).

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1991) suggest a Mesozoic–Palaeozoic crust in the Nicaraguan Highland. Unfortunately there are very few Tertiary palaeomagnetic data from the Chortis Block and a direct comparison with the data from the Nicaraguan Highland is therefore not possible.

A large counterclockwise rotation of the Chortis Block during the late Cretaceous was demonstrated by Gose (1985) and at *c.* 60 Ma the orientation of the Chortis Block was *c.* 30° different from the present orientation with respect to North America (Fig. 8)

From the development of the northern Caribbean Plate boundary, a change from northwards convergence to eastwards movement of the Caribbean Plate relative to North America in the Middle Eocene was suggested (Pindell & Dewey 1982; Sykes *et al.* 1982). This coincided with the opening of the Cayman Trough some 45–50 Myr ago (Rosencrantz *et al.* 1988). At *c.* 30 Ma there was a shift in the spreading ridge and a slow-down of the opening, which may reflect a change of the relative motion between the Caribbean and North American plates.

Our data indicate a counterclockwise rotation of the Nicaraguan Highland from *c.* 30 Ma to the present. This may be explained by left-lateral shear in the northern plate boundary related to continuing eastward motion of the Caribbean Plate and related blocks. On the basis of palaeomagnetic data from the Caribbean, most of the rotations along the plate boundaries took place in the Tertiary, commonly in post-Eocene time (Gose 1985). If the Nicaraguan Highland was part of the Chortis Block, this means that between *c.* 60 and 30 Ma this block must have rotated 60° clockwise to fit the tectonic reconstruction of the Highland. This is not supported by the trend of counterclockwise rotation for the Chortis Block during the Middle Cretaceous to Early Tertiary as suggested by Gose (1985) and may indicate that the Nicaraguan Highland acted as an independent block, separated from the Chortis Block by a prominent fault zone, for example, the Guayape Fault in southern Honduras. If this is correct the southwestern boundary of this block may coincide with the Nicaraguan Depression and the southern boundary would be located south of 12° latitude, possibly coinciding with the Hess Escarpment.

A counterclockwise rotation of the Pacific Coastal Plain since the Miocene was suggested from a palaeomagnetic study of a basaltic dyke (Sick 1989). However, palaeomagnetic poles of the present study calculated from Upper Cretaceous and Tertiary rocks of the Pacific Coastal Plain do not reveal any significant indication of rotation. The poles of the Rivas, Brito and Tamarindo formations cluster around the geographical pole. Palaeomagnetic studies in Costa Rica (Di Marco *et al.* 1995) of rocks similar to those of the Pacific Coastal Plain showed that no rotations of the Chorotega Block took place during Cretaceous–Tertiary time. From the similarity in neotectonic evolution, Weinberg (1992) suggested that northern Costa Rica and western Nicaragua were part of the same tectonic block at least since the Miocene and no major breaks separated the two areas. Di Marco *et al.* (1995) also suggested that southwestern Nicaragua forms part of the Chorotega Block.

The counterclockwise rotation of the Nicaraguan Highland ceased at *c.* 18 Ma, as indicated by the pole calculated for the Coyol Group (Table 3; Figs 7 and 8). This coincides with the time of the first deformation of the sediments of the Pacific Coastal Plain (Cruden 1989; Weinberg 1992) and the change in volcanism from the volcaniclastic Matagalpa Group to the andesitic Coyol Group. The Tamarindo Formation (*c.* 15 Myr) in the Pacific Coastal Plain is equivalent to the Coyol Group (Weyl 1980) and the calculated poles are similar but of opposite polarity (Table 3). This, in conjunction with uplift and deformation of the fore-arc sedimentary basin, may indicate accretion of the Pacific Coastal Plain and the Chorotega Block to the Nicaraguan Highland 15–18 Myr ago.

The palaeolatitudes, as calculated from the inclinations of the formation mean directions, vary. These variations may partly be due to the poor tectonic control in the Nicaraguan Highland, reflected by the sometimes large confidence cones of the mean directions, and the interpretation is therefore tentative. The calculated positions of the Highland and the Pacific Coastal Plain are in most cases around the equator or at low northern latitudes (3°S–11°N).

CONCLUSIONS

The palaeomagnetic data of this study indicate a tectonic history of the Nicaraguan Highland completely different from that of the Pacific Coastal Plain. This means that Nicaragua cannot be regarded as having been a coherent tectonic unit.

From the similarity in palaeomagnetic directions it is suggested that the Pacific Coastal Plain and the Chorotega Block were joined during the Tertiary. The change in crustal composition from continental to oceanic when moving from the Highland to the Nicaraguan Depression may support a different tectonic origin of the Highland and the Pacific Coastal Plain.

The lack of palaeomagnetic data from Tertiary rocks from Honduras or other parts of the Chortis Block prevents a direct comparison with the tectonic history of the Nicaraguan Highland. Even if the Mesozoic–Palaeozoic crust of the Chortis Block seems to continue southwards into the Nicaraguan Highland, it is still not clear if the Highland was part of the Chortis Block in the Tertiary.

Our palaeomagnetic data suggest that the Caribbean Plate in this part of Central America was not consolidated into its present shape until *c.* 15 Ma. The rotation of the Nicaraguan Highland then ceased and the Pacific Coastal Plain and the Nicaraguan Highland amalgamated.

ACKNOWLEDGMENTS

We wishes to thank G. Hodgson at the former Nicaraguan Ministry of Mines (INMINE) for geological advice and help with fieldwork. Thanks are also due to M. McWilliams for letting us use the palaeomagnetic laboratory at Stanford. This project was supported financially by the Swedish Agency for Research Cooperation with Developing Countries (SAREC).

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